

$$i12 \rightarrow$$

$$\begin{aligned} & \left(\frac{1}{6} g_2^4 y_u^{i12} [F_{3,0}[m_2] + \frac{1}{6} g_2^4 y_u^{i12} [F_{4,-1}[m_2] - \frac{8}{45} g_2^4 y_u^{i12} [F_{5,-2}[m_2] + \frac{1}{72} g_2^2 y_u^{i12} \right. \\ & \quad \left. - (18 \overline{y_e^p} y_e^p + 13 \sum p g_2^2) F_{3,0}[m^p] + \frac{1}{48} g_2^2 y_u^{i12} (12 \overline{y_e^p} y_e^p - 11 \sum p g_2^2) F_{4,-1}[m^p] + \right. \\ & \quad \left. - \frac{2}{45} g_2^4 y_u^{i12} [F_{5,-2}[m^p] + \frac{1}{24} g_2^2 y_u^{i12} (-18 (\overline{y_d^p} y_d^p + \overline{y_u^p} y_u^p) + 13 \sum p g_2^2) F_{3,0}[m^p] + \right. \\ & \quad \left. - \frac{1}{16} g_2^2 y_u^{i12} (12 \overline{y_d^p} y_d^p + 12 \overline{y_u^p} y_u^p - 11 \sum p g_2^2) F_{4,-1}[m^p] + \frac{2}{15} \sum p g_2^2 y_u^{i12} [F_{5,-2}[m^p] + \right. \\ & \quad \left. - \frac{1}{18} g_2^4 y_u^{i12} [F_{3,0}[m_d] + \frac{1}{12} g_2^4 y_u^{i12} [F_{4,-1}[\tilde{\mu}] - \frac{8}{45} g_2^4 y_u^{i12} [F_{5,-2}[\tilde{\mu}]] \right. \\ & \quad \left. - \frac{1}{6} \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{2,1,0}[m_1, m_d^r] - \frac{2}{9} g_1^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{3,1,-1}[m_1, m_d^r] + \right. \\ & \quad \left. - \frac{1}{9} g_1^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{4,1,-2}[m_1, m_d^r] + \frac{1}{36} g_1^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{2,1,0}[m_1, m_d^p] - \right. \\ & \quad \left. - \frac{1}{18} g_1^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{3,1,-1}[m_1, m_d^p] + \frac{1}{36} g_1^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{4,1,-2}[m_1, m_d^p] - \right. \\ & \quad \left. - \frac{1}{144} g_1^2 g_2^2 y_u^{i12} [F_{2,2,-1}[m_1, m_q^{i1}] + \frac{3}{4} g_2^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{2,1,0}[m_2, m_q^p] - \right. \\ & \quad \left. - \frac{3}{2} g_2^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{3,1,-1}[m_2, m_q^p] + \frac{5}{2} g_2^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{4,1,-2}[m_2, m_q^p] + \right. \\ & \quad \left. - \frac{1}{16} g_2^4 y_u^{i12} [F_{2,2,-1}[m_2, m_q^{i1}] - \frac{3}{2} g_2^4 y_u^{i12} [F_{2,1,0}[m_2, \tilde{\mu}] + \frac{1}{3} g_2^4 y_u^{i12} [F_{3,1,-1}[m_2, \tilde{\mu}] - \right. \\ & \quad \left. - \frac{5}{9} g_2^4 y_u^{i12} [F_{3,2,-2}[m_2, \tilde{\mu}] + \frac{3}{2} g_2^4 m_2^2 y_u^{i12} [F_{2,2,0}[m_2, \tilde{\mu}] + g_2^4 y_u^{i12} [F_{3,2,-3}[m_2, \tilde{\mu}] + \right. \\ & \quad \left. - g_2^4 m_2^2 y_u^{i12} [F_{3,3,-3}[m_2, \tilde{\mu}] + g_2^4 y_u^{i12} [F_{4,2,-3}[m_2, \tilde{\mu}] - g_2^4 m_2^2 y_u^{i12} [F_{3,2,-2}[m_2, \tilde{\mu}] + \right. \\ & \quad \left. - \frac{4}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{2,1,0}[m_3, m_d^r] - \frac{8}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{3,1,-1}[m_3, m_d^r] + \right. \\ & \quad \left. - \frac{4}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{4,1,-2}[m_3, m_d^r] + \frac{4}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{2,1,0}[m_3, m_q^p] - \right. \\ & \quad \left. - \frac{8}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{3,1,-1}[m_3, m_q^p] + \frac{4}{3} g_3^2 \overline{y_d^p} y_d^{i1r} y_u^{p12} [F_{4,1,-2}[m_3, m_q^p] - \right. \\ & \quad \left. - \frac{1}{3} g_2^2 g_3^2 y_u^{i12} [F_{2,2,-1}[m_3, m_q^{i1}] + g_2^2 \tilde{\mu}^2 \overline{y_e^p} y_e^p y_u^{i12} [F_{4,1,-1}[m_1^p, m_e^r] - \right. \\ & \quad \left. - \frac{1}{6} g_2^2 y_u^{i12} (\overline{a_e^p} a_e^p + 7 \tilde{\mu}^2 \overline{y_e^p} y_e^p) [F_{5,1,-2}[m_1^p, m_e^r] + 3 g_2^2 \tilde{\mu}^2 \overline{y_d^p} y_d^p y_u^{i12} \right. \\ & \quad \left. [F_{4,1,-1}[m_1^p, m_d^r] - \frac{1}{3} g_2^2 y_u^{i12} (\overline{a_d^p} a_d^p + 7 \tilde{\mu}^2 \overline{y_d^p} y_d^p) [F_{5,1,-2}[m_1^p, m_d^r] + \right. \\ & \quad \left. + \frac{3}{4} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{3,1,0}[m_q^p, m_q^s] - \frac{3}{4} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{3,1,-1}[m_q^p, m_q^s] + \right. \\ & \quad \left. + \frac{1}{4} g_2^2 y_u^{i12} (-2 \overline{a_d^p} a_d^p + \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{2,2,0}[m_q^p, m_u^r] + \frac{3}{4} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{3,1,-1}[m_q^p, m_q^s] + \right. \\ & \quad \left. - \frac{3}{4} g_2^2 y_u^{i12} (-\overline{a_d^p} a_d^p + \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{3,2,-1}[m_q^p, m_u^r] + \right. \\ & \quad \left. + \frac{1}{4} g_2^2 y_u^{i12} (\overline{a_d^p} a_d^p - 2 \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{3,2,-1}[m_q^p, m_u^r] + \right. \\ & \quad \left. + \frac{3}{4} g_2^2 y_u^{i12} (\overline{a_d^p} a_d^p - \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{4,1,-1}[m_q^p, m_u^r] + \right. \\ & \quad \left. + \frac{3}{4} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{2,1,0}[m_q^s, m_q^p] - \frac{3}{4} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{3,1,-1}[m_q^s, m_q^p] + \right. \\ & \quad \left. - \frac{1}{72} g_1^2 g_2^2 y_u^{i12} [F_{2,1,0}[m_q^{i1}, m_1] - \frac{1}{8} g_2^4 y_u^{i12} [F_{2,1,0}[m_q^{i1}, m_2] + \right. \\ & \quad \left. - \frac{2}{3} g_2^2 g_3^2 y_u^{i12} [F_{2,2,0}[m_q^{i1}, m_3] - \frac{4}{3} g_2^2 y_u^{i12} (7 \overline{a_d^p} a_d^p + \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{3,1,0}[m_u^r, m_q^p] + \right. \\ & \quad \left. + \frac{1}{2} g_2^2 y_u^{i12} (2 \overline{a_d^p} a_d^p - \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{3,2,-1}[m_u^r, m_q^p] + \right. \\ & \quad \left. - \frac{1}{2} g_2^2 y_u^{i12} (7 \overline{a_d^p} a_d^p + \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{4,1,-1}[m_u^r, m_q^p] - \right. \\ & \quad \left. - \frac{1}{2} g_2^2 y_u^{i12} (7 \overline{a_d^p} a_d^p + \tilde{\mu}^2 \overline{y_u^p} y_u^p) [F_{5,1,-2}[m_u^r, m_q^p] + \right. \\ & \quad \left. + \frac{3}{2} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{2,1,0}[m_u^t, m_d^r] - \frac{3}{2} \overline{y_d^p} y_d^s \overline{y_u^s} y_u^p y_u^{i12} [F_{3,1,-1}[m_u^t, m_d^r] + \right. \\ & \quad \left. + g_1^2 g_2^2 y_u^{i12} [F_{3,1,-1}[\tilde{\mu}, m_1] + \frac{1}{2} g_1^4 \tilde{\mu}^2 y_u^{i12} [F_{3,2,-1}[\tilde{\mu}, m_1] - \right. \\ & \quad \left. - \frac{29}{12} g_1^2 g_2^2 y_u^{i12} [F_{4,1,-2}[\tilde{\mu}, m_1] - \frac{1}{2} g_1^4 \tilde{\mu}^2 y_u^{i12} [F_{4,2,-2}[\tilde{\mu}, m_1] + \right. \\ & \quad \left. - \frac{4}{3} g_1^2 g_2^2 y_u^{i12} [F_{5,1,-3}[\tilde{\mu}, m_1] + \frac{13}{3} g_2^4 y_u^{i12} [F_{3,1,-1}[\tilde{\mu}, m_2] - \frac{4}{3} g_2^4 y_u^{i12} [F_{3,2,-2}[\tilde{\mu}, m_2] + \right. \\ & \quad \left. + \frac{1}{2} g_2^4 y_u^{i12} (4 m_2^2 + 3 \tilde{\mu}^2) [F_{3,2,-2}[\tilde{\mu}, m_2] - \frac{33}{4} g_2^4 y_u^{i12} [F_{4,1,-2}[\tilde{\mu}, m_2] + \right. \\ & \quad \left. - \frac{3}{2} g_2^4 y_u^{i1$$